

Epic 4: Model Building

Story 1: Decision Tree Model

The Decision Tree algorithm is implemented to classify flood risk based on weather-related features such as rainfall, cloud cover, temperature, and humidity. A separate function named `decisiontree()` is created to train, test, and evaluate the model using the training and testing datasets. This modular approach improves code readability, reusability, and maintainability.

Decision Tree Algorithm

The `DecisionTreeClassifier` from the Scikit-learn library is used to build the classification model. A Decision Tree is a supervised machine learning algorithm that predicts the target class by recursively splitting the dataset into decision nodes based on feature values. Each internal node represents a decision condition, while each leaf node represents the final predicted class (Flood or No Flood).

Model Training and Prediction

The model is trained using the `fit()` method, which learns the relationship between the input features (`X_train`) and the target variable (`y_train`). After training, the `predict()` method is used to generate predictions on the testing dataset (`X_test`), and the predicted values are compared with the actual labels (`y_test`) to measure model performance.

Model Evaluation

The performance of the Decision Tree model is evaluated using the following metrics:

- **Confusion Matrix** – Displays the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **Accuracy** – Measures the overall percentage of correctly classified instances.
- **Precision** – Indicates how many predicted flood events are actually correct.
- **Recall** – Measures the ability of the model to correctly identify actual flood events.
- **F1-Score** – Provides a balanced measure of Precision and Recall.
- **Support** – Represents the number of actual samples in each class.

Outcome

- Successfully trained the Decision Tree classifier.
- Generated flood predictions for the testing dataset.
- Evaluated the model using Accuracy, Confusion Matrix, Precision, Recall, and F1-Score.

- Compared the Decision Tree model with other machine learning algorithms to determine the most accurate flood prediction model.

Figure 12: Decision Tree model training and performance evaluation using the flood prediction dataset.

```
In [18]: from sklearn import tree
         from sklearn import ensemble
         from sklearn import neighbors
         import xgboost

In [32]: dtree = tree.DecisionTreeClassifier()
         Rf = ensemble.RandomForestClassifier()
         knn = neighbors.KNeighborsClassifier()
         xgb = xgboost.XGBClassifier()

In [32]: dtree.fit(x_train,y_train)
         Rf.fit(x_train,y_train)
         knn.fit(x_train,y_train)
         xgb.fit(x_train,y_train)
```